Probability and Statistics

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Lecture 6

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What we will cover today

5 - Joint Probability Distributions and Random Samples

Expected Values, Covariance, and Correlation

The Distribution of the Sample Mean

6 - Point Estimation

- Some General Concepts of Point Estimation

Expected Values, Covariance, and Correlation

Expected Values, Covariance, and Correlation

Let X and Y be jointly distributed rv's with pmf $p(x,y)$ or pdf $f(x,y)$. Then the expected value of a function $h(X,Y)$, denoted $E[h(X,Y)]$ or $\mu_{h(X,y)}$ is given by:

$$E[h(X,Y)] = \begin{cases} \sum_x \sum_y h(x,y) \cdot p(x,y) & \text{if } X,Y \text{discrete} \\ \int\limits_{-\infty}^{\infty} \int\limits_{-\infty}^{\infty} h(x,y) \cdot f(x,y) dx\, dy & \text{if } X,Y \text{continuous} \end{cases}$$

Expected Values, Covariance, and Correlation

Example:

- 5 friends have tickets to a concert
- the tickets are for seats 1-5
- tickets are randomly distributed among the five
- What is the expected number of seats separating any particular two of the five?
- possible pairs are: $(1, 2), (1, 3), \ldots, (5, 4)$

The joint pmf of (X, Y) is:

$$p(x, y) = \begin{cases} \frac{1}{20} & x = 1, \ldots, 5; y = 1, \ldots 5; x \neq y \\ 0 & \text{otherwise} \end{cases}$$

The number of seats separating the two individuals is

$$h(X, Y) = |X - Y| - 1$$

Expected Values, Covariance, and Correlation

Example:
represent $h(X,Y) = |X-Y|-1$ as a list:

$h(x, y)$		x = 1	2	3	4	5
y	1	—	0	1	2	3
	2	0	—	0	1	2
	3	1	0	—	0	1
	4	2	1	0	—	0
	5	3	2	1	0	—

Then:

$$E[h(X,Y)] = \sum_x \sum_y h(x,y) \cdot p(x,y)$$

$$= \sum_{x=1}^{5} \sum_{\substack{y=1 \\ x \neq y}}^{5} (|x-y|-1) \cdot \frac{1}{20} = 1$$

Expected Values, Covariance, and Correlation

The **covariance** between two rv's X and Y is:

Discrete X, Y

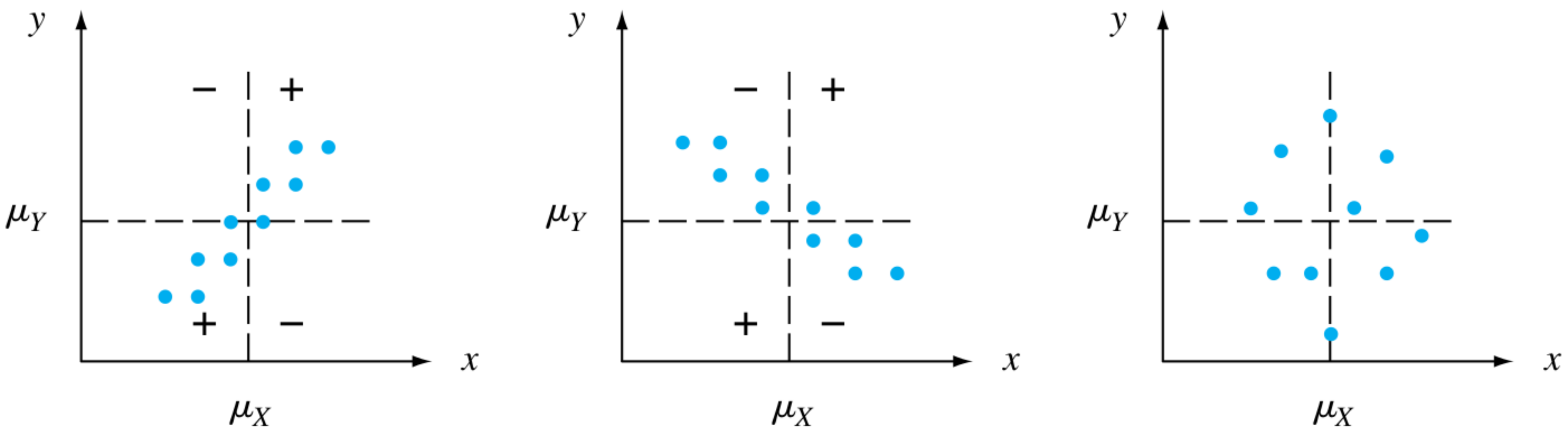
$$\begin{aligned}\text{Cov}(X,Y) &= E[(X-\mu_X)(Y-\mu_Y)] \\ &= \sum_x \sum_y (x-\mu_X)(y-\mu_Y)\,p(x,y)\end{aligned}$$

Continuous X, Y

$$\begin{aligned}\text{Cov}(X,Y) &= E[(X-\mu_X)(Y-\mu_Y)] \\ &= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} (x-\mu_X)(y-\mu_Y)\,f(x,y)\,dx\,dy\end{aligned}$$

Expected Values, Covariance, and Correlation

Graphical Explanation



- if large values of X tend to occur with large values of Y and small values of X with small values of Y (and have the same sign), then often $(x - \mu_X)$ will be large, also when $(y - \mu_Y)$ will be large (left panel): high positive covariance
- if high positive values of X appear with high negative values of y and vice versa, high negative covariance
- if there is no relation: low covariance

Expected Values, Covariance, and Correlation

Shorcut Formula for Covariance

$$\boxed{\mathsf{Cov}(X,Y) = E(XY) - E(X)\cdot E(Y)}$$

Some more properties of covariances:

$$\begin{aligned}
\mathrm{Cov}(X,X) &= \mathrm{var}(X)\\
\mathrm{Cov}(X,aY) &= a(\mathrm{Cov}(X,Y))\\
\mathrm{Cov}(X,Y+b) &= \mathrm{Cov}(X,Y)\\
\mathrm{Cov}(X,Y+Z) &= \mathrm{Cov}(X,Y) + \mathrm{Cov}(X,Z)
\end{aligned}$$

Relation to variance

$$V(X+Y) = V(X) + V(Y) + 2\,\mathrm{Cov}(X,Y)$$

Correlation

The **correlation coefficient** of X and Y, denoted $\text{Corr}(X, Y)$, $\rho_{X,Y}$ or just ρ is defined by

$$\text{Corr}(X, Y) = \rho_{X,Y} = \frac{\text{Cov}(X, Y)}{\sigma_X \cdot \sigma_Y}$$

Two properties:

1. if a and c are either both positive or both negative,

$$\text{Corr}(aX + b, cY + d) = \text{Corr}(X, Y)$$

   i.e. scaling or shifting both rv's does have an impact on their correlation coefficient

2. for any two rv's X and Y, $-1 \leq \text{Corr}(X, Y) \leq 1$

Correlation

If X and Y are independent random variables with $0 < \sigma_X < \infty$ and $0 < \sigma_Y < \infty$, then

$$\mathrm{Cov}(X,Y) = \mathrm{Corr}(X,Y) = 0.$$

Proof:
If X and Y are independent, then $E(XY) = E(X)E(Y)$ (this can be shown via marginals HW) and then

$$\mathrm{Cov}(X,Y) = E(XY) - E(X) \cdot E(Y) = 0$$

However, two dependent random variables can be uncorrelated!

Expected Values, Covariance, and Correlation

Example: Dependent but Uncorrelated Random Variables

- X can take only three values: -1, 0, 1
- each outcome has same probability
- rv Y is defined by: $Y = X^2$
- clearly these two events are dependent!

However,

$$E(XY) = E(X^3) = E(X) = 0$$

and also $E(X)E(Y) = 0$. Therefore $\mathrm{Cov}(X, Y) = 0$ and X and Y are uncorrelated.

Expected Values, Covariance, and Correlation

If Y is a linear function of X, then X and Y must be correlated, and in fact $|\,\mathrm{Corr}(X,Y)| = 1.$

If $Y = aX + b$, then $\mu_Y = a\mu_X + b$ and $Y - \mu_Y = a(X - \mu_X)$

Then:

$$\mathrm{Cov}(X,Y) = E[(X-\mu_X)(Y-\mu_Y)] = aE[(X-\mu_X)^2] = a\sigma_X^2$$

Since $\sigma_Y = |a|\sigma_X$ (because $V(Y) = V(aX+b) = a^2V(X)$)

$$\mathrm{Corr}(X,Y) = \frac{\mathrm{Cov}(X,Y)}{\sigma_X \cdot \sigma_Y} = \pm 1$$

In summary, the correlation measures the extent to which X and Y linearly related.

5 - Joint Probability Distributions and Random Samples

The Distribution of the Sample Mean

The Distribution of the Sample Mean

Let $X_1, X_2, \ldots, X_n$ be random variables, each with mean μ and variance σ^2. Let $\overline{X}_n$ be the sample mean:

$$\overline{X}_n = \frac{1}{n}(X_1 + \ldots + X_n)$$

Then the expectation of $\overline{X}_n$ is

$$E(\overline{X}_n) = \mu_{\overline{X}} = \mu$$

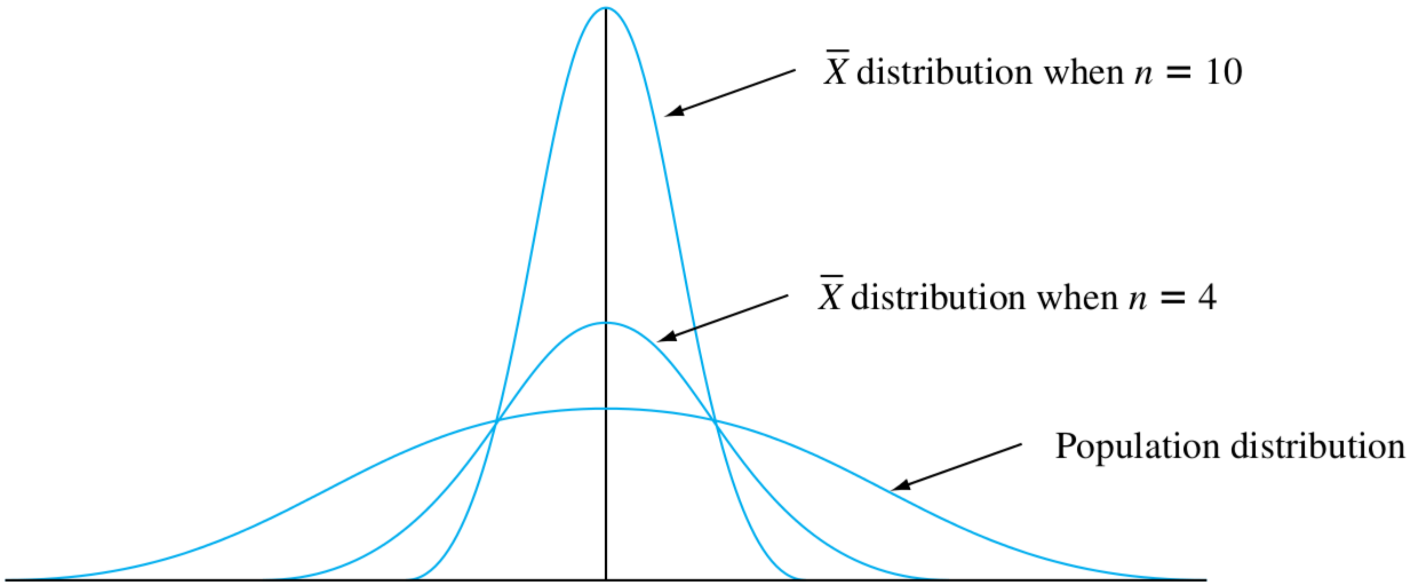
and its variance:

$$V(\overline{X}_n) = \sigma^2_{\overline{X}} = \frac{\sigma^2}{n} \qquad \text{and} \qquad \sigma_{\overline{X}} = \frac{\sigma}{\sqrt{n}}$$

- mean of $\overline{X}_n$ is equal to the mean of the distribution from which the random sample was drawn
- however, the variance of $\overline{X}_n$ is only $\frac{1}{n}$ times the variance of that distribution

The Distribution of the Sample Mean

Example for a normal distribution:



- the probability distribution of $\overline{X}_n$ is be more concentrated around the mean value μ than was the original distribution

The Distribution of the Sample Mean

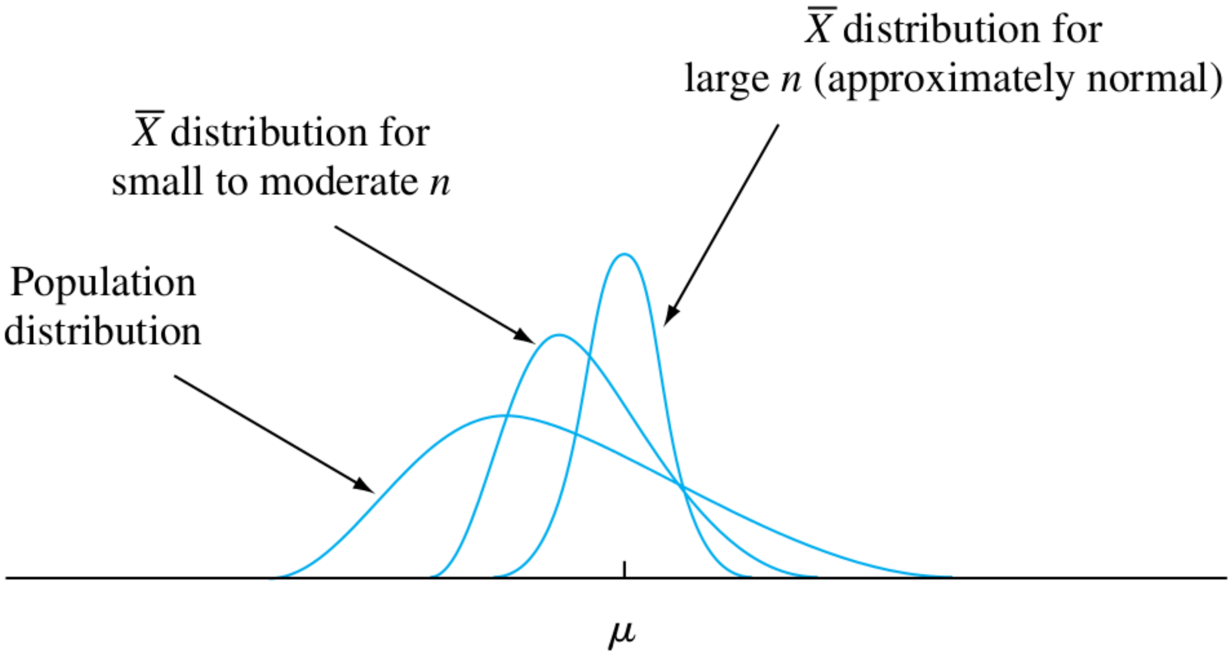
The Central Limit Theorem

Let $X_1, X_2, \ldots, X_n$ be a random sample from a distribution with mean μ and variance σ^2. Then if n is sufficiently large, $\overline{X}$ has approximately a normal distribution with $\mu_{\overline{X}} = \mu$ and $\sigma^2_{\overline{X}} = \frac{\sigma^2}{n}$. The larger the value of n, the better the approximation.

In other words:
Whenever a random sample of size n is taken from *any distribution* with mean μ and variance σ^2 , the sample average $\overline{X}$ will have a distribution that is *approximately* normal with mean μ and variance $\frac{\sigma^2}{n}$.

The Distribution of the Sample Mean

The Central Limit Theorem



- Rule of Thumb: If $n > 30$, the Central Limit Theorem can be used.

The Distribution of the Sample Mean

The Central Limit Theorem

- many random variables studied in physical experiments are approximately normal
- for example, a person's height is influenced by many random factors
- the height of each person is determined by many individual factors
- then the distribution of the heights of a large number of persons will be approximately normal

In general, the central limit theorem indicates that the distribution of the sum of many random variables can be approximately normal, even though the distribution of each random variable in the sum differs from the normal.

6 - Point Estimation

Some General Concepts of Point Estimation

Some General Concepts of Point Estimation

Example:

- a company produces lightbulbs
- and they would like to know the average lifetime of their bulbs μ
- μ is the true lifetime of the bulbs (it cannot be observed..)
- $n = 3$ with $x_1 = 5.0$, $x_2 = 6.4$ and $x_3 = 5.9$
- the computed sample mean would be $\hat{\mu} = \bar{x} = 5.77$
- this would be our "best guess" for the average lifetime

Point estimate

A point estimate of a parameter θ is a single number that can be regarded as a sensible value for θ. A point estimate is obtained by selecting a suitable statistic and computing its value from the given sample data. The selected statistic is called the point estimator of θ.

Some General Concepts of Point Estimation

Example: Car manufaturer

- the company developed a new bumber for their cars
- they are interested how well it works
- so they crash 25 cars against a wall with 20 km/h
- X = number of crashes with no damage
- parameter to be estimated: p = proportion of crashes without damage
- X is observed, $x = 15$

Then

$$\text{estimator } \hat{p} = \frac{X}{n} \qquad \text{estimate } = \frac{x}{n} = \frac{15}{25} = 0.6$$

In this case there is only one reasonable point estimator. But sometime there are more than one and we have to decide which one to choose.

Some General Concepts of Point Estimation

Example:

Let's say we have 20 observations and we want to find an estimator of the mean:

24.46	25.61	26.25	26.42	26.66	27.15	27.31	27.54	27.74	27.94
27.98	28.04	28.28	28.49	28.50	28.87	29.11	29.13	29.50	30.88

- let's assume the data comes from a normal distribution
- a normal disribution is symmetric, then the median is the mean

Here are some estimators for μ:

1. Estimator $= \overline{X}$, estimate $= \bar{x} = \sum x_i/n = 27.793$
2. Estimator $= \tilde{X}$, estimate $= \tilde{x} = (27.94 + 27.98)/2 = 27.960$
3. Estimator $= [\min(X_i) + \max(X_i)]/2$,
   estimate $= (24.46 + 30.88)/2 = 27.670$
4. Estimator = trimmed mean (discard the smallest and largest 10% of the sample and then average),
   estimate $= 27.838$

Some General Concepts of Point Estimation

Example continued:

- each estimator uses a different measure to find the center of the sample to estimate μ
- Which one is closest to the true one?
- We cannot answer this question without knowing the true value..
- ...**but** there are ways to find reasonable estimators as we will see.

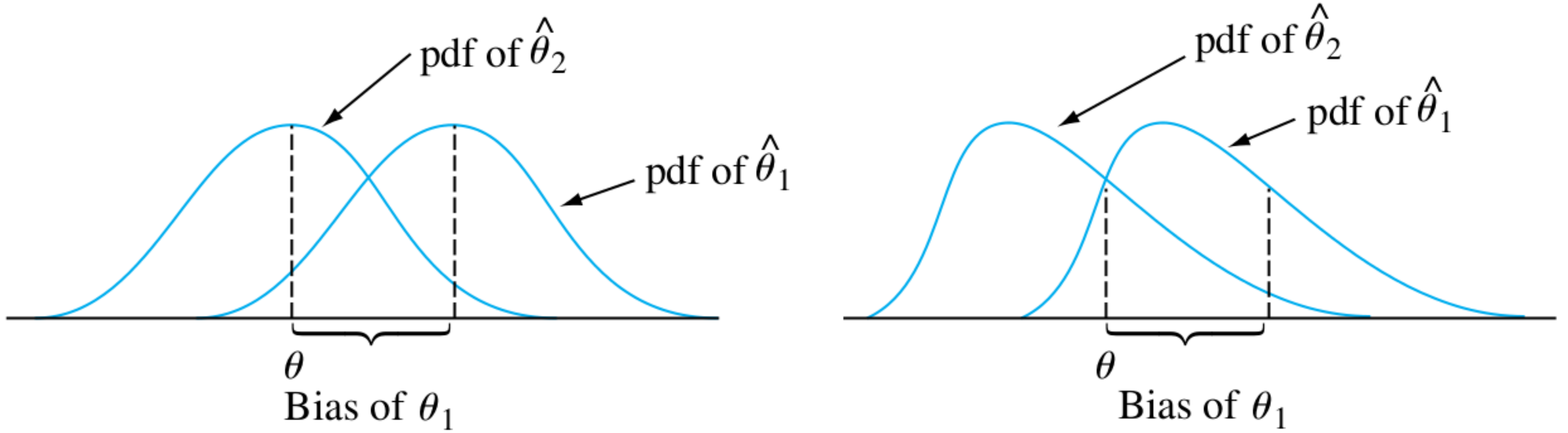
Unbiased Estimators

- suppose we have two measuring instruments
- one is accurately calibrated, but the other gives reading that are too small
- the first instrument gives measurements that are distributed about the true value (it is an unbiased estimator)
- the second gives measurements that have a systematic error (it is biased)

Some General Concepts of Point Estimation

Definition of Unbiased Estimators

A point estimator $\hat{\theta}$ is said to be an **unbiased estimator** of θ if $E(\hat{\theta}) = \theta$ for every possible value of θ. If $\hat{\theta}$ is not unbiased, the difference $E(\hat{\theta}) - \theta$ is called the **bias** of $\hat{\theta}$



The pdf's of a biased estimator θ_1 and an unbiased estimator θ_2 for a parameter θ.

Some General Concepts of Point Estimation

It may seem as though it is necessary to know the value of θ to see whether $\hat{\theta}$ is unbiased (in which case estimation is unnecessary).

However, this is usually not the case..

Example: Car crashes into wall

- we used the sample proportion X/n as an estimator of p
- where X where the number of sample successes (no damage)
- binomial distribution with parameters n and p:

$$E(\hat{p}) = E\left(\frac{X}{n}\right) = \frac{1}{n}E(X) = \frac{1}{n}(np) = p$$

> When X is a binomial rv with parameters n and p, the sample proportion $\hat{p} = X/n$ is an unbiased estimator of p.

i.e. no matter what the true value of p is, the distribution of the estimator $\hat{p}$ will be centered at the true value.

Some General Concepts of Point Estimation

Example:

- X is the reaction time to some stimulus and has uniform distribution on interval $[0 \;\; \theta]$, with θ unknown
- then the density function of X is rectangular with height $\frac{1}{\theta}$ for $0 \leq x \leq \theta$
- let's say we want to estimate θ on the basis of a random sample $X_1, X_2, \cdots, X_n$
- since θ is the largest possible time in the entire population, we choose our first estimator to be: $\hat{\theta}_1 = \max(X_1, \ldots, X_n)$
- let's say our data for $n = 5$ is: $x_1 = 4.2, \;\; x_2 = 1.7, \;\; x_3 = 2.4, \;\; x_4 = 3.9, \;\; x_5 = 1.3$
- then the point estimate of θ is: $\hat{\theta}_1 = \max(4.2, 1.7, 2.4, 3.9, 1.3) = 4.2$

Some General Concepts of Point Estimation

Example (continued):

- an unbiased estimator gives values that are sometimes greater than θ, sometimes smaller than θ
- otherwise θ could not possibly be the center (balance point) of $\hat{\theta}_1$'s distribution
- however, our proposed estimator will never overestimate θ (largest sample value cannot exceed the largest population value!)
- therefore $\hat{\theta}_1$ is a biased estimator

In fact one can show that:

$$E(\hat{\theta}_1) = \frac{n}{n+1}\theta < \theta \qquad \text{since} \qquad \frac{n}{n+1} < 1$$

The bias of $\hat{\theta}_1$ is then given by:

$$\frac{n\theta}{n+1} - \theta = \frac{n\theta - (n+1)\theta}{n+1} = \frac{n\theta - n\theta - \theta}{n+1} = \frac{-\theta}{n+1}$$

Some General Concepts of Point Estimation

Example (continued):

We can modify $\hat{\theta}_1$ to obtain an unbiased estimator:

$$\hat{\theta}_2 = \frac{n+1}{n} \max(X_1, \ldots, X_n)$$

The expectation of $\hat{\theta}_2$ is:

$$E(\hat{\theta}_2) = E\left[\frac{n+1}{n} \max(X_1, \ldots, X_n)\right] = \frac{n+1}{n} \cdot E[\max(X_1, \ldots, X_n)]$$
$$= \frac{n+1}{n} \cdot \frac{n}{n+1} \theta = \theta$$

- on different samples $\hat{\theta}_2$ will sometimes *overestimate*, sometimes *underestimate*
- but in the long run it will have *no systematic bias*

Principle of Unbiased Estimation

When choosing among several different estimators of θ, select one that is unbiased.

Some General Concepts of Point Estimation

Now let's try to find an unbiased estimator for σ^2.

Given the sample variance:

$$S^2 = \frac{(X_1 - \overline{X})^2 + (X_2 - \overline{X})^2 + \ldots + (X_n - \overline{X})^2}{n}$$
$$= \frac{\sum_{j=1}^{n}(X_j - \overline{X})^2}{n}$$

Find the expectation of S^2:

$$E[S^2] = \mu_{S^2} = \frac{n-1}{n}\sigma^2$$

Therefore $E(S^2)$ is a biased estimator of σ^2!
However, if we modify S^2 to:

$$\boxed{\hat{\sigma}^2 = \hat{S}^2 = \frac{\sum_{j=1}^{n}(X_j - \overline{X})^2}{n-1}}$$

we get an unbiased estimator of σ^2 (i.e. $E[\hat{S}^2] = \frac{n-1}{n-1}\sigma^2 = \sigma^2$).

Some General Concepts of Point Estimation

The biased estimator from our previous example:

$$E[S^2] = \mu_{S^2} = \frac{n-1}{n}\sigma^2$$

The bias is:

$$\frac{n-1}{n}\sigma^2 - \sigma^2 = \frac{n\sigma^2 - \sigma^2 - n\sigma^2}{n} = -\frac{\sigma^2}{n}$$

- the bias is negative, i.e. the estimator underestimates σ^2
- note that for large n the bias becomes small

Some General Concepts of Point Estimation

We know that:

$$E(\overline{X}) = \mu$$

so $\overline{X}$ is an unbiased estimator of μ.

In a previous example we looked at other estimators for μ:

1. Estimator $= \overline{X}$, estimate $= \bar{x} = \sum x_i/n = 27.793$
2. Estimator $= \tilde{X}$, estimate $= \tilde{x} = (27.94 + 27.98)/2 = 27.960$
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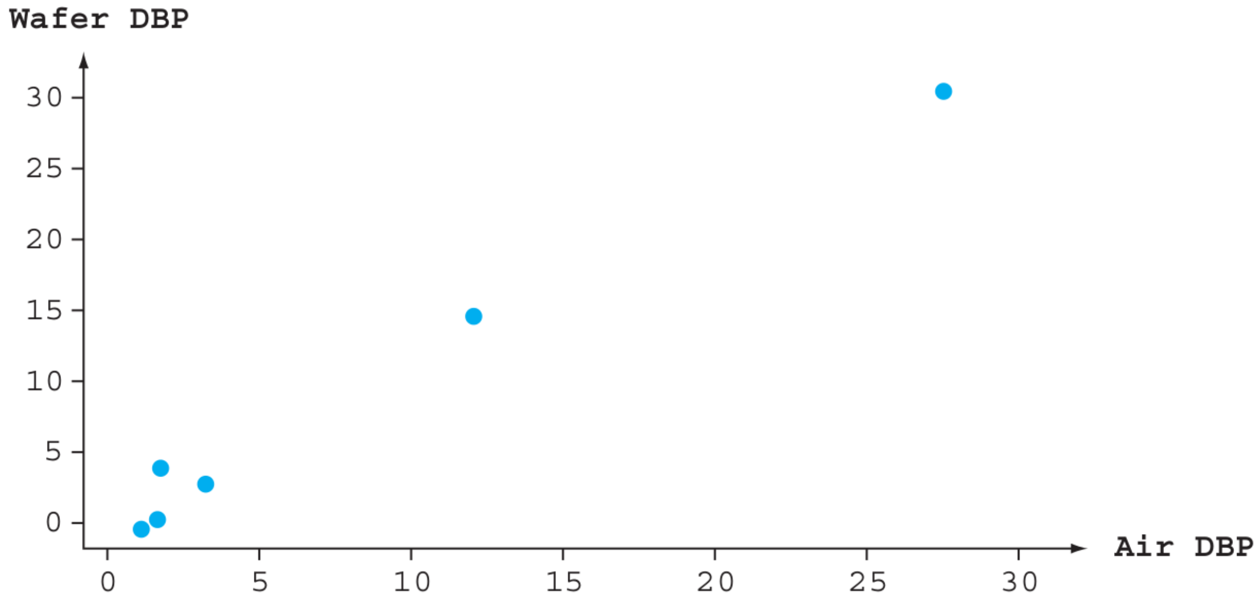
4. Estimator = trimmed mean (discard the smallest and largest 10% of the sample and then average), estimate $= 27.838$

However, if the distribution is continuous and symmetric, then $\tilde{X}$ and any trimmed mean are also unbiased estimators of μ.

Some General Concepts of Point Estimation

Example: Consider the following data:

Obs. i:	1	2	3	4	5	6
x:	.8	1.3	1.5	3.0	11.6	26.6
y:	.6	1.1	4.5	3.5	14.4	29.1



- the data looks approximately proportional
- propose $y = \beta x$ model (regression through origin)

Some General Concepts of Point Estimation

Task: estimate slope parameter β for $y = \beta x$

Consider these three possiblilities:

$$\hat{\beta} = \frac{1}{n} \sum \frac{Y_i}{x_i} \qquad \hat{\beta} = \frac{\sum Y_i}{\sum x_i} \qquad \hat{\beta} = \frac{\sum x_i Y_i}{\sum x_i^2}$$

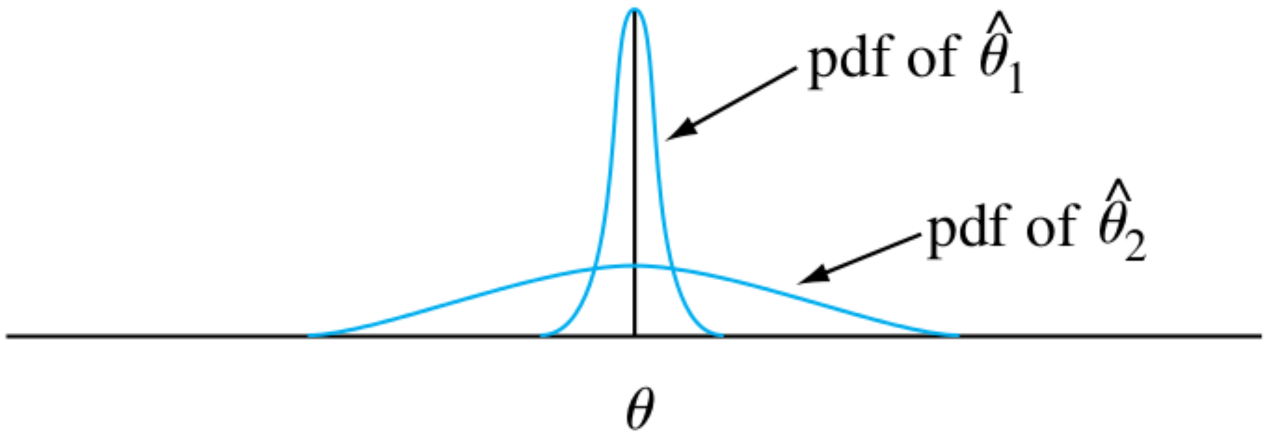
It turns out, that all three estimators are unbiased, i.e. their expectation equals β.
(This is not shown here, but can be seen in Devore p.234)

Question: How can we choose the best estimator??

Some General Concepts of Point Estimation

Estimators with Minimum Variance

Suppose $\hat{\theta}_1$ and $\hat{\theta}_2$ are two estimators of θ that are both unbiased



- the distribution of each estimator is centered at the true value of θ
- however, $\hat{\theta}_1$ is more likely than $\hat{\theta}_2$ to produce an estimate close to the true θ

Principle of **M**inimum **V**ariance **U**nbiased **E**stimation

Among all estimators of θ that are unbiased, choose the one that has minimum variance. The resulting $\hat{\theta}$ is called the **minimum variance unbiased estimator (MVUE)** of θ.

Some General Concepts of Point Estimation

In our regression example we tried to find the parameter β.

Assume that Y_i is normally distributed with mean βx_i and variance σ^2. Then it can be shown that the estimator

$$\hat{\beta} = \frac{\sum x_i Y_i}{\sum x_i^2}$$

is the **minimum variance unbiased estimator (MVUE)** for this problem.
i.e. it has a smaller variance than *any* other unbiased estimator of β.

Some General Concepts of Point Estimation

Let $X_1, X_2, \ldots, X_3$ be a random sample from a *normal distribution* with parameters μ and σ^2. Then the estimator $\hat{\mu} = \overline{X}$ is the MVUE for μ.

- whenever we believe we have a sample from a normal distribution, $\overline{X}$ should be used to estimate μ
- so in our example where we looked at four different estimators on slide 24
- the first estimator should be chosen: $\bar{x} = 27.793$